Satyajit Ghana

Engineering Leader · 3D Vision · MLOps

Bengaluru, India

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SUMMARY

Engineering leader and deep-learning engineer specializing in 3D computer vision, SLAM, and production MLOps. I design end-to-end systems — from LiDAR point-cloud reconstruction and custom neural networks to scalable cloud deployment — and lead teams shipping construction-AI scanning products. I also design and teach production MLOps curricula.

EXPERIENCE

Head of Engineering
Apr 2023 – Present

Inkers Technology Pvt Ltd, Bangalore

- Led development of new laser-scanning device prototypes with improved laser accuracy and integrated RGB and thermal cameras.
- Developed a modified SLAM algorithm that enables scanning in small, confined spaces and reduced drift by 50%.
- Built a point-cloud cleaning algorithm that reduced cloud thickness from 2 cm to 1 cm while preserving detail.

Stack | ROS1, C++, PCL, Eigen, Python, Open3D, CloudCompare

MLOps Instructor
Feb 2022 – Present

The School of AI, Bangalore

- Designed and taught the EMLOV2 and EMLOV3 production-MLOps courses end to end.
- Curriculum spanned Docker & Compose, PyTorch Lightning, DVC, experiment tracking, HPO, Gradio, FastAPI + Redis serving, TorchServe, in-browser models, Kubernetes/EKS, ISTIO, KServe, Kubeflow Pipelines, canary deployments, Prometheus & Grafana, PEFT/LoRA, vLLM, Stable Diffusion/SDXL, LLM fine-tuning, GGML & llama.cpp, and LlamaIndex & LangChain.

Stack | PyTorch, HuggingFace, Docker, AWS (EC2/S3/ECR/EKS/ECS/EFS), Kubernetes, TorchServe, GGML

Deep Learning Software Engineer
Apr 2022 – May 2023

Inkers Technology Pvt Ltd, Bangalore

- Optimized the point-cloud cleaning and meshing algorithm to produce meshes accurate to within 1 cm.
- Built a construction-defect identification and tracking module on 3D textured meshes by projecting segmentation into 3D and clustering.
- Designed and trained a custom transformer for 3D mesh part segmentation, deriving construction material quantities.
- Reduced mesh sizes from 1.2 GB to 120 MB via texture optimization, compression, and Blender decimation, enabling in-browser rendering.
- Migrated the Observance web app from Bootstrap to TailwindCSS + PrimeReact and built a task-management module with roles and authorization.
- Synchronized LiDAR, camera, and IMU streams to improve the 3D SLAM algorithm and reduce drift; cut GPU load 30% with on-demand rendering and computed floor-plan deviations with Hausdorff distance on the frontend.
- Built a one-of-a-kind thermal-camera integration with 3D scanning to produce 3D thermal maps, visualized on the frontend.

Stack | PCL, ROS, Open3D, Blender, PyTorch, TailwindCSS, React, THREE.js, MeshLab, CloudCompare

Associate Deep Learning Software Engineer
Jul 2021 – Apr 2022

Inkers Technology Pvt Ltd, Bangalore

- Developed the full-stack web app for Observance.AI, an end-to-end construction-AI platform.
- Migrated facial-recognition onboarding to AWS Lambda, onboarding people in minutes with up to 2,000 parallel Lambdas.
- Built a 3D LiDAR scan visualization platform with annotations, area/length/angle/volume measurement, synced renderers, and post-processing.
- Optimized the data pipeline on HPC instances (p3/p4, g4dn, r6i), processing 1–2 TB/day on S3 with 128-core / 1024 GB spot instances.
- Implemented 3D reconstruction and texturing from LiDAR + RGB SLAM, automating the pipeline with Blender scripting for structural-defect identification.
- Developed a camera-to-LiDAR texture optimization algorithm in PyTorch3D that reduced photometric projection error.

Stack | React, Flutter, DevExtreme, THREE.js, Frappe, PyTorch, LiDAR, AWS (EC2/S3/ECR), Kinesis Video Streams

Visual Processing Intern
Jun 2020 – Jul 2021

Inkers Technology Pvt Ltd, Bangalore

- Built a driver-monitoring system estimating driver state from glucose levels, pose, and eye landmarks, with a real-time Flutter app receiving Jetson Nano updates over Bluetooth.
- Designed a custom GStreamer plugin for DeepStream performing near-real-time stereo rectification, and deployed on edge devices (Xavier NX) via Docker at stereo 4K @ 30 fps for detection and tracking.
- Optimized YOLOv4-Tiny for TensorRT for a 2x speedup and managed multi-GPU training on AWS (8x V100) with data backup and synchronization to cut costs.
- Built a synthetic dataset generator robust to camera glitches, motion blur, and object scale — expanding 100–200 source images to 80K and detecting objects as small as 10x10 px.

Stack | DeepStream, GStreamer, TensorRT, OpenCV, Docker, Flutter, AWS, PyTorch

SKILLS

Languages | Python, C/C++, JavaScript, TypeScript, Java, Haskell, Go

ML & Deep Learning | PyTorch, PyTorch3D, HuggingFace, TensorRT, DeepStream, vLLM, PEFT/LoRA, Stable Diffusion/SDXL, LlamaIndex, LangChain

3D & Vision | SLAM, 3D Reconstruction, Point Clouds (PCL, Open3D, CloudCompare, MeshLab), Blender (BPY), THREE.js, OpenCV

MLOps & Cloud | Docker, Kubernetes (EKS/ECS), KServe, Kubeflow, TorchServe, ISTIO, AWS (Lambda, SageMaker, S3, Kinesis, EC2, ECR, EFS), Azure, Prometheus, Grafana

Backend & Systems | FastAPI, gRPC, Redis, Kafka, RabbitMQ, WebRTC, ROS1, Hydra

Frontend & Mobile | React, TailwindCSS, Flutter

EDUCATION

B.Tech, Computer Science & Engineering
Aug 2017 – Mar 2021

Ramaiah University of Applied Sciences, Bangalore — *CGPA 9.0/10*

Relevant coursework: Object-Oriented Programming, Databases, Discrete Mathematics, Data Structures & Algorithms, Operating Systems, Computer Networks, Machine Learning, Data Mining, Advanced Data Structures & Algorithms, Information Retrieval, Image Processing.